

Project Executable Files

Date:	15 March 2026
Team ID:	(xxxx)
Project Name:	Human Development Index (HDI) Prediction Using Machine Learning
Maximum Marks:	3 Marks

Project Executable Files

The **Human Development Index (HDI) Prediction System** is developed using **Python, Flask, Scikit-learn, Pandas, NumPy, HTML, CSS, JavaScript, and Visual Studio Code**. The project predicts the HDI category of a country based on **Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI Per Capita**. All required source code, model files, datasets, and documentation are organized for easy execution and evaluation.

Step 1: Submission Checklist:

S.No	Item to Submit	Submitted
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide	Yes
3	requirements.txt (dependency file)	Yes
4	Database schema / seed files (Not Applicable – CSV dataset used)	NA
5	Environment configuration file (.env.example)	NA
6	Deployed application URL	No (Add if deployed)
7	Executable binary / APK	NA
8	Dockerfile / Containerization configuration	NA
9	Test files and test results	Yes

S.No	Item to Submit	Submitted
10	Demo video / Walkthrough	Yes (If required by your mentor)

Step 2: File / Folder Structure:-

HDI-Prediction/

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└─ dataset/

|

└─ hdi_dataset.csv

|

└─ models/

|

└─ model.pkl

|

└─ static/

|

└─ style.css

|

└─ script.js

|

└─ templates/

|

└─ index.html

|

└─ app.py

└─ train_model.py

└─ requirements.txt

└─ README.md

└─ test_prediction.py

|

└─ .gitignore

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	<i>Add your deployed application URL (if hosted on Render, Railway, or PythonAnywhere).</i>
Login Credentials (Demo)	Not Required (The application does not require login.)
Platform / Hosting Provider	Local Flask Server (Development). Future deployment can use Render, Railway, or PythonAnywhere.
Repository Link	<i>Paste your GitHub repository URL here.</i>
Demo Video Link	<i>Paste your demo video link here (Google Drive/YouTube, if required).</i>

Step 4: Run Instructions:-

Pre-requisites

- Windows / Linux / macOS
- Python 3.x
- Visual Studio Code
- Flask
- Pandas
- NumPy
- Scikit-learn
- Joblib

Steps

1. Clone the repository:
2. `git clone <GitHub Repository URL>`
3. Open the project in **Visual Studio Code**.
4. Install all required dependencies:
5. `pip install -r requirements.txt`

6. Verify that hdi_dataset.csv and model.pkl are available in the appropriate folders.
7. Start the Flask application:
8. `python app.py`
9. Open a web browser and visit:
10. `http://127.0.0.1:5000`
11. Enter:
 - Life Expectancy
 - Mean Years of Schooling
 - Expected Years of Schooling
 - GNI Per Capita
12. Click the **Predict** button to view the predicted HDI category.

Step 5: Known Issues / Limitations:-

Known Issue / Limitation	Workaround / Status
Uses a local dataset only	Can be enhanced by integrating live UNDP data in future versions.
Local Flask server is intended for development	Deploy on a production server such as Render or Railway for wider access.
Prediction depends on the quality of the training dataset	Improve accuracy by expanding and updating the dataset regularly.